TABLE 7. THE χ^2-DISTRIBUTION FUNCTION

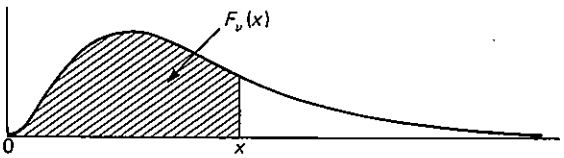
The function tabulated is

$$F_\nu(x) = \frac{1}{2^{\nu/2}\Gamma(\frac{\nu}{2})} \int_0^x t^{\frac{1}{2}\nu-1} e^{-\frac{1}{2}t}\, dt$$

for integer $\nu \leqslant 25$. $F_\nu(x)$ is the probability that a random variable X, distributed as χ^2 with ν degrees of freedom, will be less than or equal to x. Note that $F_1(x) = 2\Phi(x^{\frac{1}{2}}) - 1$ (cf. Table 4). For certain values of x and $\nu > 25$ use may be made of the following relation between the χ^2- and Poisson distributions:

$$F_\nu(x) = 1 - F(\tfrac{1}{2}\nu - 1 | \tfrac{1}{2}x)$$

where $F(r|\mu)$ is the Poisson distribution function (see Table 2). If $\nu > 25$, X is approximately normally distributed



(The above shape applies for $\nu \geqslant 3$ only. When $\nu < 3$ the mode is at the origin.)

with mean ν and variance 2ν. A better approximation is usually obtained by using the formula

$$F_\nu(x) \doteqdot \Phi(\sqrt{2x} - \sqrt{2\nu - 1})$$

where $\Phi(s)$ is the normal distribution function (see Table 4).

Omitted entries to the left and right of tabulated values are 1 and 0 respectively (to four decimal places).

$\nu =$	1	$\nu =$	1	$\nu =$	2	$\nu =$	2	$\nu =$	3	$\nu =$	3
$x = 0{\cdot}0$	0·0000	$x = 4{\cdot}0$	0·9545	$x = 0{\cdot}0$	0·0000	$x = 4{\cdot}0$	0·8647	$x = 0{\cdot}0$	0·0000	$x = 4{\cdot}0$	0·7385
·1	·2482	·1	·9571	·1	·0488	·1	·8713	·1	·0082	·2	·7593
·2	·3453	·2	·9596	·2	·0952	·2	·8775	·2	·0224	·4	·7786
·3	·4161	·3	·9619	·3	·1393	·3	·8835	·3	·0400	·6	·7965
·4	·4729	·4	·9641	·4	·1813	·4	·8892	·4	·0598	·8	·8130
0·5	0·5205	4·5	0·9661	0·5	0·2212	4·5	0·8946	0·5	0·0811	5·0	0·8282
·6	·5614	·6	·9680	·6	·2592	·6	·8997	·6	·1036	·2	·8423
·7	·5972	·7	·9698	·7	·2953	·7	·9046	·7	·1268	·4	·8553
·8	·6289	·8	·9715	·8	·3297	·8	·9093	·8	·1505	·6	·8672
·9	·6572	·9	·9731	·9	·3624	·9	·9137	·9	·1746	·8	·8782
1·0	0·6827	5·0	0·9747	1·0	0·3935	5·0	0·9179	1·0	0·1987	6·0	0·8884
·1	·7057	·1	·9761	·1	·4231	·1	·9219	·1	·2229	·2	·8977
·2	·7267	·2	·9774	·2	·4512	·2	·9257	·2	·2470	·4	·9063
·3	·7458	·3	·9787	·3	·4780	·3	·9293	·3	·2709	·6	·9142
·4	·7633	·4	·9799	·4	·5034	·4	·9328	·4	·2945	·8	·9214
1·5	0·7793	5·5	0·9810	1·5	0·5276	5·5	0·9361	1·5	0·3177	7·0	0·9281
·6	·7941	·6	·9820	·6	·5507	·6	·9392	·6	·3406	·2	·9342
·7	·8077	·7	·9830	·7	·5726	·7	·9422	·7	·3631	·4	·9398
·8	·8203	·8	·9840	·8	·5934	·8	·9450	·8	·3851	·6	·9450
·9	·8319	·9	·9849	·9	·6133	·9	·9477	·9	·4066	·8	·9497
2·0	0·8427	6·0	0·9857	2·0	0·6321	6·0	0·9502	2·0	0·4276	8·0	0·9540
·1	·8527	·1	·9865	·1	·6501	·2	·9550	·1	·4481	·2	·9579
·2	·8620	·2	·9872	·2	·6671	·4	·9592	·2	·4681	·4	·9616
·3	·8706	·3	·9879	·3	·6834	·6	·9631	·3	·4875	·6	·9649
·4	·8787	·4	·9886	·4	·6988	·8	·9666	·4	·5064	·8	·9679
2·5	0·8862	6·5	0·9892	2·5	0·7135	7·0	0·9698	2·5	0·5247	9·0	0·9707
·6	·8931	·6	·9898	·6	·7275	·2	·9727	·6	·5425	·2	·9733
·7	·8997	·7	·9904	·7	·7408	·4	·9753	·7	·5598	·4	·9756
·8	·9057	·8	·9909	·8	·7534	·6	·9776	·8	·5765	·6	·9777
·9	·9114	·9	·9914	·9	·7654	·8	·9798	·9	·5927	·8	·9797
3·0	0·9167	7·0	0·9918	3·0	0·7769	8·0	0·9817	3·0	0·6084	10·0	0·9814
·1	·9217	·1	·9923	·1	·7878	·2	·9834	·1	·6235	·2	·9831
·2	·9264	·2	·9927	·2	·7981	·4	·9850	·2	·6382	·4	·9845
·3	·9307	·3	·9931	·3	·8080	·6	·9864	·3	·6524	·6	·9859
·4	·9348	·4	·9935	·4	·8173	·8	·9877	·4	·6660	·8	·9871
3·5	0·9386	7·5	0·9938	3·5	0·8262	9·0	0·9889	3·5	0·6792	11·0	0·9883
·6	·9422	·6	·9942	·6	·8347	·2	·9899	·6	·6920	·2	·9893
·7	·9456	·7	·9945	·7	·8428	·4	·9909	·7	·7043	·4	·9903
·8	·9487	·8	·9948	·8	·8504	·6	·9918	·8	·7161	·6	·9911
·9	·9517	·9	·9951	·9	·8577	·8	·9926	·9	·7275	·8	·9919
4·0	0·9545	8·0	0·9953	4·0	0·8647	10·0	0·9933	4·0	0·7385	12·0	0·9926

TABLE 7. THE χ^2-DISTRIBUTION FUNCTION

$\nu =$	4	5	6	7	8	9	10	11	12	13	14
$x = 0.5$	0.0265	0.0079	0.0022	0.0006	0.0001						
1.0	.0902	.0374	.0144	.0052	.0018	0.0006	0.0002	0.0001			
1.5	.1734	.0869	.0405	.0177	.0073	.0029	.0011	.0004	0.0001		
2.0	.2642	.1509	.0803	.0402	.0190	.0085	.0037	.0015	.0006	0.0002	0.0001
2.5	0.3554	0.2235	0.1315	0.0729	0.0383	0.0191	0.0091	0.0042	0.0018	0.0008	0.0003
3.0	.4422	.3000	.1912	.1150	.0656	.0357	.0186	.0093	.0045	.0021	.0009
3.5	.5221	.3766	.2560	.1648	.1008	.0589	.0329	.0177	.0091	.0046	.0022
4.0	.5940	.4506	.3233	.2202	.1429	.0886	.0527	.0301	.0166	.0088	.0045
4.5	.6575	.5201	.3907	.2793	.1906	.1245	.0780	.0471	.0274	.0154	.0084
5.0	0.7127	0.5841	0.4562	0.3400	0.2424	0.1657	0.1088	0.0688	0.0420	0.0248	0.0142
5.5	.7603	.6421	.5185	.4008	.2970	.2113	.1446	.0954	.0608	.0375	.0224
6.0	.8009	.6938	.5768	.4603	.3528	.2601	.1847	.1266	.0839	.0538	.0335
6.5	.8352	.7394	.6304	.5173	.4086	.3110	.2283	.1620	.1112	.0739	.0477
7.0	.8641	.7794	.6792	.5711	.4634	.3629	.2746	.2009	.1424	.0978	.0653
7.5	0.8883	0.8140	0.7229	0.6213	0.5162	0.4148	0.3225	0.2427	0.1771	0.1254	0.0863
8.0	.9084	.8438	.7619	.6674	.5665	.4659	.3712	.2867	.2149	.1564	.1107
8.5	.9251	.8693	.7963	.7094	.6138	.5154	.4199	.3321	.2551	.1904	.1383
9.0	.9389	.8909	.8264	.7473	.6577	.5627	.4679	.3781	.2971	.2271	.1689
9.5	.9503	.9093	.8527	.7813	.6981	.6075	.5146	.4242	.3403	.2658	.2022
10.0	0.9596	0.9248	0.8753	0.8114	0.7350	0.6495	0.5595	0.4696	0.3840	0.3061	0.2378
10.5	.9672	.9378	.8949	.8380	.7683	.6885	.6022	.5140	.4278	.3474	.2752
11.0	.9734	.9486	.9116	.8614	.7983	.7243	.6425	.5567	.4711	.3892	.3140
11.5	.9785	.9577	.9259	.8818	.8251	.7570	.6801	.5976	.5134	.4310	.3536
12.0	.9826	.9652	.9380	.8994	.8488	.7867	.7149	.6364	.5543	.4724	.3937
12.5	0.9860	0.9715	0.9483	0.9147	0.8697	0.8134	0.7470	0.6727	0.5936	0.5129	0.4338
13.0	.9887	.9766	.9570	.9279	.8882	.8374	.7763	.7067	.6310	.5522	.4735
13.5	.9909	.9809	.9643	.9392	.9042	.8587	.8030	.7381	.6662	.5900	.5124
14.0	.9927	.9844	.9704	.9488	.9182	.8777	.8270	.7670	.6993	.6262	.5503
14.5	.9941	.9873	.9755	.9570	.9304	.8944	.8486	.7935	.7301	.6604	.5868
15.0	0.9953	0.9896	0.9797	0.9640	0.9409	0.9091	0.8679	0.8175	0.7586	0.6926	0.6218
15.5	.9962	.9916	.9833	.9699	.9499	.9219	.8851	.8393	.7848	.7228	.6551
16.0	.9970	.9932	.9862	.9749	.9576	.9331	.9004	.8589	.8088	.7509	.6866
16.5	.9976	.9944	.9887	.9791	.9642	.9429	.9138	.8764	.8306	.7768	.7162
17.0	.9981	.9955	.9907	.9826	.9699	.9513	.9256	.8921	.8504	.8007	.7438
17.5	0.9985	0.9964	0.9924	0.9856	0.9747	0.9586	0.9360	0.9061	0.8683	0.8226	0.7695
18.0	.9988	.9971	.9938	.9880	.9788	.9648	.9450	.9184	.8843	.8425	.7932
18.5	.9990	.9976	.9949	.9901	.9822	.9702	.9529	.9293	.8987	.8606	.8151
19.0	.9992	.9981	.9958	.9918	.9851	.9748	.9597	.9389	.9115	.8769	.8351
19.5	.9994	.9984	.9966	.9932	.9876	.9787	.9656	.9473	.9228	.8916	.8533
20	0.9995	0.9988	0.9972	0.9944	0.9897	0.9821	0.9707	0.9547	0.9329	0.9048	0.8699
21	.9997	.9992	.9982	.9962	.9929	.9873	.9789	.9666	.9496	.9271	.8984
22	.9998	.9995	.9988	.9975	.9951	.9911	.9849	.9756	.9625	.9446	.9214
23	.9999	.9997	.9992	.9983	.9966	.9938	.9893	.9823	.9723	.9583	.9397
24	.9999	.9998	.9995	.9989	.9977	.9957	.9924	.9873	.9797	.9689	.9542
25	0.9999	0.9999	0.9997	0.9992	0.9984	0.9970	0.9947	0.9909	0.9852	0.9769	0.9654
26		.9999	.9998	.9995	.9989	.9980	.9963	.9935	.9893	.9830	.9741
27		.9999	.9999	.9997	.9993	.9986	.9974	.9954	.9923	.9876	.9807
28			.9999	.9998	.9995	.9990	.9982	.9968	.9945	.9910	.9858
29			.9999	.9999	.9997	.9994	.9988	.9977	.9961	.9935	.9895
30				0.9999	0.9998	0.9996	0.9991	0.9984	0.9972	0.9953	0.9924

TABLE 7. THE χ^2-DISTRIBUTION FUNCTION

$\nu =$	15	16	17	18	19	20	21	22	23	24	25
$\varkappa = 3$	0·0004	0·0002	0·0001								
4	·0023	·0011	·0005	0·0002	0·0001						
5	0·0079	0·0042	0·0022	0·0011	0·0006	0·0003	0·0001	0·0001			
6	·0203	·0119	·0068	·0038	·0021	·0011	·0006	·0003	0·0001	0·0001	
7	·0424	·0267	·0165	·0099	·0058	·0033	·0019	·0010	·0005	·0003	0·0001
8	·0762	·0511	·0335	·0214	·0133	·0081	·0049	·0028	·0016	·0009	·0005
9	·1225	·0866	·0597	·0403	·0265	·0171	·0108	·0067	·0040	·0024	·0014
10	0·1803	0·1334	0·0964	0·0681	0·0471	0·0318	0·0211	0·0137	0·0087	0·0055	0·0033
11	·2474	·1905	·1434	·1056	·0762	·0538	·0372	·0253	·0168	·0110	·0071
12	·3210	·2560	·1999	·1528	·1144	·0839	·0604	·0426	·0295	·0201	·0134
13	·3977	·3272	·2638	·2084	·1614	·1226	·0914	·0668	·0480	·0339	·0235
14	·4745	·4013	·3329	·2709	·2163	·1695	·1304	·0985	·0731	·0533	·0383
15	0·5486	0·4754	0·4045	0·3380	0·2774	0·2236	0·1770	0·1378	0·1054	0·0792	0·0586
16	·6179	·5470	·4762	·4075	·3427	·2834	·2303	·1841	·1447	·1119	·0852
17	·6811	·6144	·5456	·4769	·4101	·3470	·2889	·2366	·1907	·1513	·1182
18	·7373	·6761	·6112	·5443	·4776	·4126	·3510	·2940	·2425	·1970	·1576
19	·7863	·7313	·6715	·6082	·5432	·4782	·4149	·3547	·2988	·2480	·2029
20	0·8281	0·7798	0·7258	0·6672	0·6054	0·5421	0·4787	0·4170	0·3581	0·3032	0·2532
21	·8632	·8215	·7737	·7206	·6632	·6029	·5411	·4793	·4189	·3613	·3074
22	·8922	·8568	·8153	·7680	·7157	·6595	·6005	·5401	·4797	·4207	·3643
23	·9159	·8863	·8507	·8094	·7627	·7112	·6560	·5983	·5392	·4802	·4224
24	·9349	·9105	·8806	·8450	·8038	·7576	·7069	·6528	·5962	·5384	·4806
25	0·9501	0·9302	0·9053	0·8751	0·8395	0·7986	0·7528	0·7029	0·6497	0·5942	0·5376
26	·9620	·9460	·9255	·9002	·8698	·8342	·7936	·7483	·6991	·6468	·5924
27	·9713	·9585	·9419	·9210	·8953	·8647	·8291	·7888	·7440	·6955	·6441
28	·9784	·9684	·9551	·9379	·9166	·8906	·8598	·8243	·7842	·7400	·6921
29	·9839	·9761	·9655	·9516	·9340	·9122	·8860	·8551	·8197	·7799	·7361
30	0·9881	0·9820	0·9737	0·9626	0·9482	0·9301	0·9080	0·8815	0·8506	0·8152	0·7757
31	·9912	·9865	·9800	·9712	·9596	·9448	·9263	·9039	·8772	·8462	·8110
32	·9936	·9900	·9850	·9780	·9687	·9567	·9414	·9226	·8999	·8730	·8420
33	·9953	·9926	·9887	·9833	·9760	·9663	·9538	·9381	·9189	·8959	·8689
34	·9966	·9946	·9916	·9874	·9816	·9739	·9638	·9509	·9348	·9153	·8921
35	0·9975	0·9960	0·9938	0·9905	0·9860	0·9799	0·9718	0·9613	0·9480	0·9316	0·9118
36	·9982	·9971	·9954	·9929	·9894	·9846	·9781	·9696	·9587	·9451	·9284
37	·9987	·9979	·9966	·9948	·9921	·9883	·9832	·9763	·9675	·9562	·9423
38	·9991	·9985	·9975	·9961	·9941	·9911	·9871	·9817	·9745	·9653	·9537
39	·9994	·9989	·9982	·9972	·9956	·9933	·9902	·9859	·9802	·9727	·9632
40	0·9995	0·9992	0·9987	0·9979	0·9967	0·9950	0·9926	0·9892	0·9846	0·9786	0·9708
41	·9997	·9994	·9991	·9985	·9976	·9963	·9944	·9918	·9882	·9833	·9770
42	·9998	·9996	·9993	·9989	·9982	·9972	·9958	·9937	·9909	·9871	·9820
43	·9998	·9997	·9995	·9992	·9987	·9980	·9969	·9953	·9931	·9901	·9860
44	·9999	·9998	·9997	·9994	·9991	·9985	·9977	·9965	·9947	·9924	·9892
45	0·9999	0·9999	0·9998	0·9996	0·9993	0·9989	0·9983	0·9973	0·9960	0·9942	0·9916
46	·9999	·9999	·9998	·9997	·9995	·9992	·9987	·9980	·9970	·9956	·9936
47		·9999	·9999	·9998	·9996	·9994	·9991	·9985	·9978	·9967	·9951
48			·9999	·9998	·9997	·9996	·9993	·9989	·9983	·9975	·9963
49			·9999	·9999	·9998	·9997	·9995	·9992	·9988	·9981	·9972
50				0·9999	0·9999	0·9998	0·9996	0·9994	0·9991	0·9986	0·9979

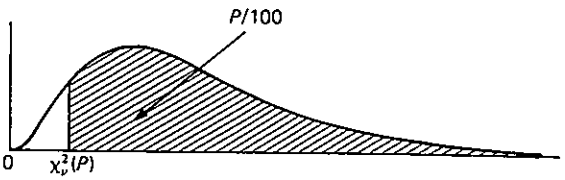
TABLE 8. PERCENTAGE POINTS OF THE χ^2-DISTRIBUTION

This table gives percentage points $\chi^2_\nu(P)$ defined by the equation

$$\frac{P}{100} = \frac{1}{2^{\nu/2}\,\Gamma(\frac{\nu}{2})} \int_{\chi^2_\nu(P)}^{\infty} x^{\frac{1}{2}\nu-1} e^{-\frac{1}{2}x}\, dx.$$

If X is a variable distributed as χ^2 with ν degrees of freedom, $P/100$ is the probability that $X \geqslant \chi^2_\nu(P)$.

For $\nu > 100$, $\sqrt{2X}$ is approximately normally distributed with mean $\sqrt{2\nu-1}$ and unit variance.



(The above shape applies for $\nu \geqslant 3$ only. When $\nu < 3$ the mode is at the origin.)

P	99·95	99·9	99·5	99	97·5	95	90	80	70	60
$\nu = 1$	$0{\cdot}0^6 3927$	$0{\cdot}0^5 1571$	$0{\cdot}0^4 3927$	$0{\cdot}0^3 1571$	$0{\cdot}0^3 9821$	0·003932	0·01579	0·06418	0·1485	0·2750
2	0·001000	0·002001	0·01003	0·02010	0·05064	0·1026	0·2107	0·4463	0·7133	1·022
3	0·01528	0·02430	0·07172	0·1148	0·2158	0·3518	0·5844	1·005	1·424	1·869
4	0·06392	0·09080	0·2070	0·2971	0·4844	0·7107	1·064	1·649	2·195	2·753
5	0·1581	0·2102	0·4117	0·5543	0·8312	1·145	1·610	2·343	3·000	3·655
6	0·2994	0·3811	0·6757	0·8721	1·237	1·635	2·204	3·070	3·828	4·570
7	0·4849	0·5985	0·9893	1·239	1·690	2·167	2·833	3·822	4·671	5·493
8	0·7104	0·8571	1·344	1·646	2·180	2·733	3·490	4·594	5·527	6·423
9	0·9717	1·152	1·735	2·088	2·700	3·325	4·168	5·380	6·393	7·357
10	1·265	1·479	2·156	2·558	3·247	3·940	4·865	6·179	7·267	8·295
11	1·587	1·834	2·603	3·053	3·816	4·575	5·578	6·989	8·148	9·237
12	1·934	2·214	3·074	3·571	4·404	5·226	6·304	7·807	9·034	10·18
13	2·305	2·617	3·565	4·107	5·009	5·892	7·042	8·634	9·926	11·13
14	2·697	3·041	4·075	4·660	5·629	6·571	7·790	9·467	10·82	12·08
15	3·108	3·483	4·601	5·229	6·262	7·261	8·547	10·31	11·72	13·03
16	3·536	3·942	5·142	5·812	6·908	7·962	9·312	11·15	12·62	13·98
17	3·980	4·416	5·697	6·408	7·564	8·672	10·09	12·00	13·53	14·94
18	4·439	4·905	6·265	7·015	8·231	9·390	10·86	12·86	14·44	15·89
19	4·912	5·407	6·844	7·633	8·907	10·12	11·65	13·72	15·35	16·85
20	5·398	5·921	7·434	8·260	9·591	10·85	12·44	14·58	16·27	17·81
21	5·896	6·447	8·034	8·897	10·28	11·59	13·24	15·44	17·18	18·77
22	6·404	6·983	8·643	9·542	10·98	12·34	14·04	16·31	18·10	19·73
23	6·924	7·529	9·260	10·20	11·69	13·09	14·85	17·19	19·02	20·69
24	7·453	8·085	9·886	10·86	12·40	13·85	15·66	18·06	19·94	21·65
25	7·991	8·649	10·52	11·52	13·12	14·61	16·47	18·94	20·87	22·62
26	8·538	9·222	11·16	12·20	13·84	15·38	17·29	19·82	21·79	23·58
27	9·093	9·803	11·81	12·88	14·57	16·15	18·11	20·70	22·72	24·54
28	9·656	10·39	12·46	13·56	15·31	16·93	18·94	21·59	23·65	25·51
29	10·23	10·99	13·12	14·26	16·05	17·71	19·77	22·48	24·58	26·48
30	10·80	11·59	13·79	14·95	16·79	18·49	20·60	23·36	25·51	27·44
32	11·98	12·81	15·13	16·36	18·29	20·07	22·27	25·15	27·37	29·38
34	13·18	14·06	16·50	17·79	19·81	21·66	23·95	26·94	29·24	31·31
36	14·40	15·32	17·89	19·23	21·34	23·27	25·64	28·73	31·12	33·25
38	15·64	16·61	19·29	20·69	22·88	24·88	27·34	30·54	32·99	35·19
40	16·91	17·92	20·71	22·16	24·43	26·51	29·05	32·34	34·87	37·13
50	23·46	24·67	27·99	29·71	32·36	34·76	37·69	41·45	44·31	46·86
60	30·34	31·74	35·53	37·48	40·48	43·19	46·46	50·64	53·81	56·62
70	37·47	39·04	43·28	45·44	48·76	51·74	55·33	59·90	63·35	66·40
80	44·79	46·52	51·17	53·54	57·15	60·39	64·28	69·21	72·92	76·19
90	52·28	54·16	59·20	61·75	65·65	69·13	73·29	78·56	82·51	85·99
100	59·90	61·92	67·33	70·06	74·22	77·93	82·36	87·95	92·13	95·81

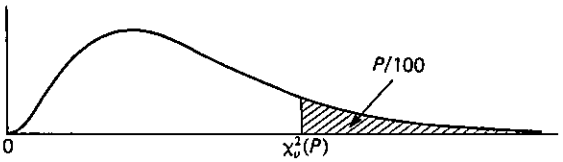
TABLE 8. PERCENTAGE POINTS OF THE χ^2-DISTRIBUTION

This table gives percentage points $\chi^2_\nu(P)$ defined by the equation

$$\frac{P}{100} = \frac{1}{2^{\nu/2}\,\Gamma(\frac{\nu}{2})} \int_{\chi^2_\nu(P)}^{\infty} x^{\frac{1}{2}\nu-1}\, e^{-\frac{1}{2}x}\, dx.$$

If X is a variable distributed as χ^2 with ν degrees of freedom, $P/100$ is the probability that $X \geqslant \chi^2_\nu(P)$.

For $\nu > 100$, $\sqrt{2X}$ is approximately normally distributed with mean $\sqrt{2\nu-1}$ and unit variance.



(The above shape applies for $\nu \geqslant 3$ only. When $\nu < 3$ the mode is at the origin.)

P	50	40	30	20	10	5	2·5	1	0·5	0·1	0·05
$\nu = 1$	0·4549	0·7083	1·074	1·642	2·706	3·841	5·024	6·635	7·879	10·83	12·12
2	1·386	1·833	2·408	3·219	4·605	5·991	7·378	9·210	10·60	13·82	15·20
3	2·366	2·946	3·665	4·642	6·251	7·815	9·348	11·34	12·84	16·27	17·73
4	3·357	4·045	4·878	5·989	7·779	9·488	11·14	13·28	14·86	18·47	20·00
5	4·351	5·132	6·064	7·289	9·236	11·07	12·83	15·09	16·75	20·52	22·11
6	5·348	6·211	7·231	8·558	10·64	12·59	14·45	16·81	18·55	22·46	24·10
7	6·346	7·283	8·383	9·803	12·02	14·07	16·01	18·48	20·28	24·32	26·02
8	7·344	8·351	9·524	11·03	13·36	15·51	17·53	20·09	21·95	26·12	27·87
9	8·343	9·414	10·66	12·24	14·68	16·92	19·02	21·67	23·59	27·88	29·67
10	9·342	10·47	11·78	13·44	15·99	18·31	20·48	23·21	25·19	29·59	31·42
11	10·34	11·53	12·90	14·63	17·28	19·68	21·92	24·72	26·76	31·26	33·14
12	11·34	12·58	14·01	15·81	18·55	21·03	23·34	26·22	28·30	32·91	34·82
13	12·34	13·64	15·12	16·98	19·81	22·36	24·74	27·69	29·82	34·53	36·48
14	13·34	14·69	16·22	18·15	21·06	23·68	26·12	29·14	31·32	36·12	38·11
15	14·34	15·73	17·32	19·31	22·31	25·00	27·49	30·58	32·80	37·70	39·72
16	15·34	16·78	18·42	20·47	23·54	26·30	28·85	32·00	34·27	39·25	41·31
17	16·34	17·82	19·51	21·61	24·77	27·59	30·19	33·41	35·72	40·79	42·88
18	17·34	18·87	20·60	22·76	25·99	28·87	31·53	34·81	37·16	42·31	44·43
19	18·34	19·91	21·69	23·90	27·20	30·14	32·85	36·19	38·58	43·82	45·97
20	19·34	20·95	22·77	25·04	28·41	31·41	34·17	37·57	40·00	45·31	47·50
21	20·34	21·99	23·86	26·17	29·62	32·67	35·48	38·93	41·40	46·80	49·01
22	21·34	23·03	24·94	27·30	30·81	33·92	36·78	40·29	42·80	48·27	50·51
23	22·34	24·07	26·02	28·43	32·01	35·17	38·08	41·64	44·18	49·73	52·00
24	23·34	25·11	27·10	29·55	33·20	36·42	39·36	42·98	45·56	51·18	53·48
25	24·34	26·14	28·17	30·68	34·38	37·65	40·65	44·31	46·93	52·62	54·95
26	25·34	27·18	29·25	31·79	35·56	38·89	41·92	45·64	48·29	54·05	56·41
27	26·34	28·21	30·32	32·91	36·74	40·11	43·19	46·96	49·64	55·48	57·86
28	27·34	29·25	31·39	34·03	37·92	41·34	44·46	48·28	50·99	56·89	59·30
29	28·34	30·28	32·46	35·14	39·09	42·56	45·72	49·59	52·34	58·30	60·73
30	29·34	31·32	33·53	36·25	40·26	43·77	46·98	50·89	53·67	59·70	62·16
32	31·34	33·38	35·66	38·47	42·58	46·19	49·48	53·49	56·33	62·49	65·00
34	33·34	35·44	37·80	40·68	44·90	48·60	51·97	56·06	58·96	65·25	67·80
36	35·34	37·50	39·92	42·88	47·21	51·00	54·44	58·62	61·58	67·99	70·59
38	37·34	39·56	42·05	45·08	49·51	53·38	56·90	61·16	64·18	70·70	73·35
40	39·34	41·62	44·16	47·27	51·81	55·76	59·34	63·69	66·77	73·40	76·09
50	49·33	51·89	54·72	58·16	63·17	67·50	71·42	76·15	79·49	86·66	89·56
60	59·33	62·13	65·23	68·97	74·40	79·08	83·30	88·38	91·95	99·61	102·7
70	69·33	72·36	75·69	79·71	85·53	90·53	95·02	100·4	104·2	112·3	115·6
80	79·33	82·57	86·12	90·41	96·58	101·9	106·6	112·3	116·3	124·8	128·3
90	89·33	92·76	96·52	101·1	107·6	113·1	118·1	124·1	128·3	137·2	140·8
100	99·33	102·9	106·9	111·7	118·5	124·3	129·6	135·8	140·2	149·4	153·2

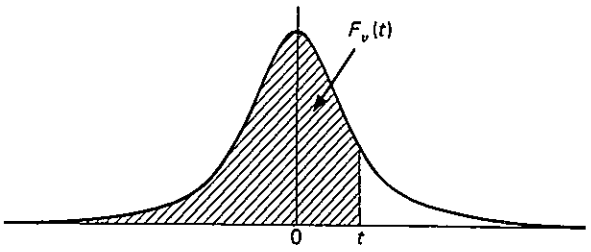
TABLE 9. THE t-DISTRIBUTION FUNCTION

The function tabulated is

$$F_\nu(t) = \frac{1}{\sqrt{\nu\pi}} \frac{\Gamma(\frac{1}{2}\nu+\frac{1}{2})}{\Gamma(\frac{1}{2}\nu)} \int_{-\infty}^{t} \frac{ds}{(1+s^2/\nu)^{\frac{1}{2}(\nu+1)}}.$$

$F_\nu(t)$ is the probability that a random variable, distributed as t with ν degrees of freedom, will be less than or equal to t. When $t < 0$ use $F_\nu(t) = 1 - F_\nu(-t)$, the t distribution being symmetric about zero.

The limiting distribution of t as ν tends to infinity is the normal distribution with zero mean and unit variance (see Table 4). When ν is large interpolation in ν should be harmonic.



Omitted entries to the right of tabulated values are 1 (to four decimal places).

$\nu =$	1	$\nu =$	1	$\nu =$	2	$\nu =$	2	$\nu =$	3	$\nu =$	3
$t = 0.0$	0.5000	$t = 4.0$	0.9220	$t = 0.0$	0.5000	$t = 4.0$	0.9714	$t = 0.0$	0.5000	$t = 4.0$	0.9860
.1	.5317	4.2	.9256	.1	.5353	.1	.9727	.1	.5367	.1	.9869
.2	.5628	4.4	.9289	.2	.5700	.2	.9739	.2	.5729	.2	.9877
.3	.5928	4.6	.9319	.3	.6038	.3	.9750	.3	.6081	.3	.9884
.4	.6211	4.8	.9346	.4	.6361	.4	.9760	.4	.6420	.4	.9891
0.5	0.6476	5.0	0.9372	0.5	0.6667	4.5	0.9770	0.5	0.6743	4.5	0.9898
.6	.6720	5.5	.9428	.6	.6953	.6	.9779	.6	.7046	.6	.9903
.7	.6944	6.0	.9474	.7	.7218	.7	.9788	.7	.7328	.7	.9909
.8	.7148	6.5	.9514	.8	.7462	.8	.9796	.8	.7589	.8	.9914
.9	.7333	7.0	.9548	.9	.7684	.9	.9804	.9	.7828	.9	.9919
1.0	0.7500	7.5	0.9578	1.0	0.7887	5.0	0.9811	1.0	0.8045	5.0	0.9923
.1	.7651	8.0	.9604	.1	.8070	.1	.9818	.1	.8242	.1	.9927
.2	.7789	8.5	.9627	.2	.8235	.2	.9825	.2	.8419	.2	.9931
.3	.7913	9.0	.9648	.3	.8384	.3	.9831	.3	.8578	.3	.9934
.4	.8026	9.5	.9666	.4	.8518	.4	.9837	.4	.8720	.4	.9938
1.5	0.8128	10.0	0.9683	1.5	0.8638	5.5	0.9842	1.5	0.8847	5.5	0.9941
.6	.8222	10.5	.9698	.6	.8746	.6	.9848	.6	.8960	.6	.9944
.7	.8307	11.0	.9711	.7	.8844	.7	.9853	.7	.9062	.7	.9946
.8	.8386	11.5	.9724	.8	.8932	.8	.9858	.8	.9152	.8	.9949
.9	.8458	12.0	.9735	.9	.9011	.9	.9862	.9	.9232	.9	.9951
2.0	0.8524	12.5	0.9746	2.0	0.9082	6.0	0.9867	2.0	0.9303	6.0	0.9954
.1	.8585	13.0	.9756	.1	.9147	.1	.9871	.1	.9367	.1	.9956
.2	.8642	13.5	.9765	.2	.9206	.2	.9875	.2	.9424	.2	.9958
.3	.8695	14.0	.9773	.3	.9259	.3	.9879	.3	.9475	.3	.9960
.4	.8743	14.5	.9781	.4	.9308	.4	.9882	.4	.9521	.4	.9961
2.5	0.8789	15	0.9788	2.5	0.9352	6.5	0.9886	2.5	0.9561	6.5	0.9963
.6	.8831	16	.9801	.6	.9392	.6	.9889	.6	.9598	.6	.9965
.7	.8871	17	.9813	.7	.9429	.7	.9892	.7	.9631	.7	.9966
.8	.8908	18	.9823	.8	.9463	.8	.9895	.8	.9661	.8	.9967
.9	.8943	19	.9833	.9	.9494	.9	.9898	.9	.9687	.9	.9969
3.0	0.8976	20	0.9841	3.0	0.9523	7.0	0.9901	3.0	0.9712	7.0	0.9970
.1	.9007	21	.9849	.1	.9549	.1	.9904	.1	.9734	.1	.9971
.2	.9036	22	.9855	.2	.9573	.2	.9906	.2	.9753	.2	.9972
.3	.9063	23	.9862	.3	.9596	.3	.9909	.3	.9771	.3	.9973
.4	.9089	24	.9867	.4	.9617	.4	.9911	.4	.9788	.4	.9974
3.5	0.9114	25	0.9873	3.5	0.9636	7.5	0.9913	3.5	0.9803	7.5	0.9975
.6	.9138	30	.9894	.6	.9654	.6	.9916	.6	.9816	.6	.9976
.7	.9160	35	.9909	.7	.9670	.7	.9918	.7	.9829	.7	.9977
.8	.9181	40	.9920	.8	.9686	.8	.9920	.8	.9840	.8	.9978
.9	.9201	45	.9929	.9	.9701	.9	.9922	.9	.9850	.9	.9979
4.0	0.9220	50	0.9936	4.0	0.9714	8.0	0.9924	4.0	0.9860	8.0	0.9980

TABLE 9. THE *t*-DISTRIBUTION FUNCTION

$\nu =$	4	5	6	7	8	9	10	11	12	13	14
$t = 0.0$	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
.1	.5374	.5379	.5382	.5384	.5386	.5387	.5388	.5389	.5390	.5391	.5391
.2	.5744	.5753	.5760	.5764	.5768	.5770	.5773	.5774	.5776	.5777	.5778
.3	.6104	.6119	.6129	.6136	.6141	.6145	.6148	.6151	.6153	.6155	.6157
.4	.6452	.6472	.6485	.6495	.6502	.6508	.6512	.6516	.6519	.6522	.6524
0.5	0.6783	0.6809	0.6826	0.6838	0.6847	0.6855	0.6861	0.6865	0.6869	0.6873	0.6876
.6	.7096	.7127	.7148	.7163	.7174	.7183	.7191	.7197	.7202	.7206	.7210
.7	.7387	.7424	.7449	.7467	.7481	.7492	.7501	.7508	.7514	.7519	.7523
.8	.7657	.7700	.7729	.7750	.7766	.7778	.7788	.7797	.7804	.7810	.7815
.9	.7905	.7953	.7986	.8010	.8028	.8042	.8054	.8063	.8071	.8078	.8083
1.0	0.8130	0.8184	0.8220	0.8247	0.8267	0.8283	0.8296	0.8306	0.8315	0.8322	0.8329
.1	.8335	.8393	.8433	.8461	.8483	.8501	.8514	.8526	.8535	.8544	.8551
.2	.8518	.8581	.8623	.8654	.8678	.8696	.8711	.8723	.8734	.8742	.8750
.3	.8683	.8748	.8793	.8826	.8851	.8870	.8886	.8899	.8910	.8919	.8927
.4	.8829	.8898	.8945	.8979	.9005	.9025	.9041	.9055	.9066	.9075	.9084
1.5	0.8960	0.9030	0.9079	0.9114	0.9140	0.9161	0.9177	0.9191	0.9203	0.9212	0.9221
.6	.9076	.9148	.9196	.9232	.9259	.9280	.9297	.9310	.9322	.9332	.9340
.7	.9178	.9251	.9300	.9335	.9362	.9383	.9400	.9414	.9426	.9435	.9444
.8	.9269	.9341	.9390	.9426	.9452	.9473	.9490	.9503	.9515	.9525	.9533
.9	.9349	.9421	.9469	.9504	.9530	.9551	.9567	.9580	.9591	.9601	.9609
2.0	0.9419	0.9490	0.9538	0.9572	0.9597	0.9617	0.9633	0.9646	0.9657	0.9666	0.9674
.1	.9482	.9551	.9598	.9631	.9655	.9674	.9690	.9702	.9712	.9721	.9728
.2	.9537	.9605	.9649	.9681	.9705	.9723	.9738	.9750	.9759	.9768	.9774
.3	.9585	.9651	.9694	.9725	.9748	.9765	.9779	.9790	.9799	.9807	.9813
.4	.9628	.9692	.9734	.9763	.9784	.9801	.9813	.9824	.9832	.9840	.9846
2.5	0.9666	0.9728	0.9767	0.9795	0.9815	0.9831	0.9843	0.9852	0.9860	0.9867	0.9873
.6	.9700	.9759	.9797	.9823	.9842	.9856	.9868	.9877	.9884	.9890	.9895
.7	.9730	.9786	.9822	.9847	.9865	.9878	.9888	.9897	.9903	.9909	.9914
.8	.9756	.9810	.9844	.9867	.9884	.9896	.9906	.9914	.9920	.9925	.9929
.9	.9779	.9831	.9863	.9885	.9901	.9912	.9921	.9928	.9933	.9938	.9942
3.0	0.9800	0.9850	0.9880	0.9900	0.9915	0.9925	0.9933	0.9940	0.9945	0.9949	0.9952
.1	.9819	.9866	.9894	.9913	.9927	.9936	.9944	.9949	.9954	.9958	.9961
.2	.9835	.9880	.9907	.9925	.9937	.9946	.9953	.9958	.9962	.9965	.9968
.3	.9850	.9893	.9918	.9934	.9946	.9954	.9960	.9965	.9968	.9971	.9974
.4	.9864	.9904	.9928	.9943	.9953	.9961	.9966	.9970	.9974	.9976	.9978
3.5	0.9876	0.9914	0.9936	0.9950	0.9960	0.9966	0.9971	0.9975	0.9978	0.9980	0.9982
.6	.9886	.9922	.9943	.9956	.9965	.9971	.9976	.9979	.9982	.9984	.9986
.7	.9896	.9930	.9950	.9962	.9970	.9975	.9979	.9982	.9985	.9987	.9988
.8	.9904	.9937	.9955	.9966	.9974	.9979	.9983	.9985	.9987	.9989	.9990
.9	.9912	.9943	.9960	.9971	.9977	.9982	.9985	.9988	.9989	.9991	.9992
4.0	0.9919	0.9948	0.9964	0.9974	0.9980	0.9984	0.9987	0.9990	0.9991	0.9992	0.9993
.1	.9926	.9953	.9968	.9977	.9983	.9987	.9989	.9991	.9993	.9994	.9995
.2	.9932	.9958	.9972	.9980	.9985	.9988	.9991	.9993	.9994	.9995	.9996
.3	.9937	.9961	.9975	.9982	.9987	.9990	.9992	.9994	.9995	.9996	.9996
.4	.9942	.9965	.9977	.9984	.9989	.9991	.9993	.9995	.9996	.9996	.9997
4.5	0.9946	0.9968	0.9979	0.9986	0.9990	0.9993	0.9994	0.9995	0.9996	0.9997	0.9998
.6	.9950	.9971	.9982	.9988	.9991	.9994	.9995	.9996	.9997	.9998	.9998
.7	.9953	.9973	.9983	.9989	.9992	.9994	.9996	.9997	.9997	.9998	.9998
.8	.9957	.9976	.9985	.9990	.9993	.9995	.9996	.9997	.9998	.9998	.9999
.9	.9960	.9978	.9986	.9991	.9994	.9996	.9997	.9998	.9998	.9999	.9999
5.0	0.9963	0.9979	0.9988	0.9992	0.9995	0.9996	0.9997	0.9998	0.9998	0.9999	0.9999

TABLE 9. THE t-DISTRIBUTION FUNCTION

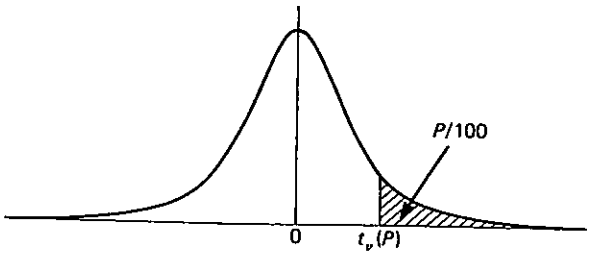
$\nu =$	15	16	17	18	19	20	24	30	40	60	∞
$t = 0.0$	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
.1	.5392	.5392	.5392	.5393	.5393	.5393	.5394	.5395	.5396	.5397	.5398
.2	.5779	.5780	.5781	.5781	.5782	.5782	.5784	.5786	.5788	.5789	.5793
.3	.6159	.6160	.6161	.6162	.6163	.6164	.6166	.6169	.6171	.6174	.6179
.4	.6526	.6528	.6529	.6531	.6532	.6533	.6537	.6540	.6544	.6547	.6554
0.5	0.6878	0.6881	0.6883	0.6884	0.6886	0.6887	0.6892	0.6896	0.6901	0.6905	0.6915
.6	.7213	.7215	.7218	.7220	.7222	.7224	.7229	.7235	.7241	.7246	.7257
.7	.7527	.7530	.7533	.7536	.7538	.7540	.7547	.7553	.7560	.7567	.7580
.8	.7819	.7823	.7826	.7829	.7832	.7834	.7842	.7850	.7858	.7866	.7881
.9	.8088	.8093	.8097	.8100	.8103	.8106	.8115	.8124	.8132	.8141	.8159
1.0	0.8334	0.8339	0.8343	0.8347	0.8351	0.8354	0.8364	0.8373	0.8383	0.8393	0.8413
.1	.8557	.8562	.8567	.8571	.8575	.8578	.8589	.8600	.8610	.8621	.8643
.2	.8756	.8762	.8767	.8772	.8776	.8779	.8791	.8802	.8814	.8826	.8849
.3	.8934	.8940	.8945	.8950	.8954	.8958	.8970	.8982	.8995	.9007	.9032
.4	.9091	.9097	.9103	.9107	.9112	.9116	.9128	.9141	.9154	.9167	.9192
1.5	0.9228	0.9235	0.9240	0.9245	0.9250	0.9254	0.9267	0.9280	0.9293	0.9306	0.9332
.6	.9348	.9354	.9360	.9365	.9370	.9374	.9387	.9400	.9413	.9426	.9452
.7	.9451	.9458	.9463	.9468	.9473	.9477	.9490	.9503	.9516	.9528	.9554
.8	.9540	.9546	.9552	.9557	.9561	.9565	.9578	.9590	.9603	.9616	.9641
.9	.9616	.9622	.9627	.9632	.9636	.9640	.9652	.9665	.9677	.9689	.9713
2.0	0.9680	0.9686	0.9691	0.9696	0.9700	0.9704	0.9715	0.9727	0.9738	0.9750	0.9772
.1	.9735	.9740	.9745	.9750	.9753	.9757	.9768	.9779	.9790	.9800	.9821
.2	.9781	.9786	.9790	.9794	.9798	.9801	.9812	.9822	.9832	.9842	.9861
.3	.9819	.9824	.9828	.9832	.9835	.9838	.9848	.9857	.9866	.9875	.9893
.4	.9851	.9855	.9859	.9863	.9866	.9869	.9877	.9886	.9894	.9902	.9918
2.5	0.9877	0.9882	0.9885	0.9888	0.9891	0.9894	0.9902	0.9909	0.9917	0.9924	0.9938
.6	.9900	.9903	.9907	.9910	.9912	.9914	.9921	.9928	.9935	.9941	.9953
.7	.9918	.9921	.9924	.9927	.9929	.9931	.9937	.9944	.9949	.9955	.9965
.8	.9933	.9936	.9938	.9941	.9943	.9945	.9950	.9956	.9961	.9966	.9974
.9	.9945	.9948	.9950	.9952	.9954	.9956	.9961	.9965	.9970	.9974	.9981
3.0	0.9955	0.9958	0.9960	0.9962	0.9963	0.9965	0.9969	0.9973	0.9977	0.9980	0.9987
.1	.9963	.9966	.9967	.9969	.9971	.9972	.9976	.9979	.9982	.9985	.9990
.2	.9970	.9972	.9974	.9975	.9976	.9978	.9981	.9984	.9987	.9989	.9993
.3	.9976	.9977	.9979	.9980	.9981	.9982	.9985	.9988	.9990	.9992	.9995
.4	.9980	.9982	.9983	.9984	.9985	.9986	.9988	.9990	.9992	.9994	.9997
3.5	0.9984	0.9985	0.9986	0.9987	0.9988	0.9989	0.9991	0.9993	0.9994	0.9996	0.9998
.6	.9987	.9988	.9989	.9990	.9990	.9991	.9993	.9994	.9996	.9997	.9998
.7	.9989	.9990	.9991	.9992	.9992	.9993	.9994	.9996	.9997	.9998	.9999
.8	.9991	.9992	.9993	.9993	.9994	.9994	.9996	.9997	.9998	.9998	.9999
.9	.9993	.9994	.9994	.9995	.9995	.9996	.9997	.9997	.9998	.9999	
4.0	0.9994	0.9995	0.9995	0.9996	0.9996	0.9996	0.9997	0.9998	0.9999	0.9999	
.1	.9995	.9996	.9996	.9997	.9997	.9997	.9998	.9999	.9999	.9999	
.2	.9996	.9997	.9997	.9997	.9998	.9998	.9998	.9999	.9999		
.3	.9997	.9997	.9998	.9998	.9998	.9998	.9999	.9999	.9999		
.4	.9997	.9998	.9998	.9998	.9998	.9999	.9999	.9999			
4.5	0.9998	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999				

TABLE 10. PERCENTAGE POINTS OF THE t-DISTRIBUTION

This table gives percentage points $t_\nu(P)$ defined by the equation

$$\frac{P}{100} = \frac{1}{\sqrt{\nu\pi}} \frac{\Gamma(\frac{1}{2}\nu+\frac{1}{2})}{\Gamma(\frac{1}{2}\nu)} \int_{t_\nu(P)}^{\infty} \frac{dt}{(1+t^2/\nu)^{\frac{1}{2}(\nu+1)}}.$$

Let X_1 and X_2 be independent random variables having a normal distribution with zero mean and unit variance and a χ^2-distribution with ν degrees of freedom respectively; then $t = X_1/\sqrt{X_2/\nu}$ has Student's t-distribution with ν degrees of freedom, and the probability that $t \geqslant t_\nu(P)$ is $P/100$. The lower percentage points are given by symmetry as $-t_\nu(P)$, and the probability that $|t| \geqslant t_\nu(P)$ is $2P/100$.



The limiting distribution of t as ν tends to infinity is the normal distribution with zero mean and unit variance. When ν is large interpolation in ν should be harmonic.

P	40	30	25	20	15	10	5	2·5	1	0·5	0·1	0·05
$\nu = 1$	0·3249	0·7265	1·0000	1·3764	1·963	3·078	6·314	12·71	31·82	63·66	318·3	636·6
2	0·2887	0·6172	0·8165	1·0607	1·386	1·886	2·920	4·303	6·965	9·925	22·33	31·60
3	0·2767	0·5844	0·7649	0·9785	1·250	1·638	2·353	3·182	4·541	5·841	10·21	12·92
4	0·2707	0·5686	0·7407	0·9410	1·190	1·533	2·132	2·776	3·747	4·604	7·173	8·610
5	0·2672	0·5594	0·7267	0·9195	1·156	1·476	2·015	2·571	3·365	4·032	5·893	6·869
6	0·2648	0·5534	0·7176	0·9057	1·134	1·440	1·943	2·447	3·143	3·707	5·208	5·959
7	0·2632	0·5491	0·7111	0·8960	1·119	1·415	1·895	2·365	2·998	3·499	4·785	5·408
8	0·2619	0·5459	0·7064	0·8889	1·108	1·397	1·860	2·306	2·896	3·355	4·501	5·041
9	0·2610	0·5435	0·7027	0·8834	1·100	1·383	1·833	2·262	2·821	3·250	4·297	4·781
10	0·2602	0·5415	0·6998	0·8791	1·093	1·372	1·812	2·228	2·764	3·169	4·144	4·587
11	0·2596	0·5399	0·6974	0·8755	1·088	1·363	1·796	2·201	2·718	3·106	4·025	4·437
12	0·2590	0·5386	0·6955	0·8726	1·083	1·356	1·782	2·179	2·681	3·055	3·930	4·318
13	0·2586	0·5375	0·6938	0·8702	1·079	1·350	1·771	2·160	2·650	3·012	3·852	4·221
14	0·2582	0·5366	0·6924	0·8681	1·076	1·345	1·761	2·145	2·624	2·977	3·787	4·140
15	0·2579	0·5357	0·6912	0·8662	1·074	1·341	1·753	2·131	2·602	2·947	3·733	4·073
16	0·2576	0·5350	0·6901	0·8647	1·071	1·337	1·746	2·120	2·583	2·921	3·686	4·015
17	0·2573	0·5344	0·6892	0·8633	1·069	1·333	1·740	2·110	2·567	2·898	3·646	3·965
18	0·2571	0·5338	0·6884	0·8620	1·067	1·330	1·734	2·101	2·552	2·878	3·610	3·922
19	0·2569	0·5333	0·6876	0·8610	1·066	1·328	1·729	2·093	2·539	2·861	3·579	3·883
20	0·2567	0·5329	0·6870	0·8600	1·064	1·325	1·725	2·086	2·528	2·845	3·552	3·850
21	0·2566	0·5325	0·6864	0·8591	1·063	1·323	1·721	2·080	2·518	2·831	3·527	3·819
22	0·2564	0·5321	0·6858	0·8583	1·061	1·321	1·717	2·074	2·508	2·819	3·505	3·792
23	0·2563	0·5317	0·6853	0·8575	1·060	1·319	1·714	2·069	2·500	2·807	3·485	3·768
24	0·2562	0·5314	0·6848	0·8569	1·059	1·318	1·711	2·064	2·492	2·797	3·467	3·745
25	0·2561	0·5312	0·6844	0·8562	1·058	1·316	1·708	2·060	2·485	2·787	3·450	3·725
26	0·2560	0·5309	0·6840	0·8557	1·058	1·315	1·706	2·056	2·479	2·779	3·435	3·707
27	0·2559	0·5306	0·6837	0·8551	1·057	1·314	1·703	2·052	2·473	2·771	3·421	3·690
28	0·2558	0·5304	0·6834	0·8546	1·056	1·313	1·701	2·048	2·467	2·763	3·408	3·674
29	0·2557	0·5302	0·6830	0·8542	1·055	1·311	1·699	2·045	2·462	2·756	3·396	3·659
30	0·2556	0·5300	0·6828	0·8538	1·055	1·310	1·697	2·042	2·457	2·750	3·385	3·646
32	0·2555	0·5297	0·6822	0·8530	1·054	1·309	1·694	2·037	2·449	2·738	3·365	3·622
34	0·2553	0·5294	0·6818	0·8523	1·052	1·307	1·691	2·032	2·441	2·728	3·348	3·601
36	0·2552	0·5291	0·6814	0·8517	1·052	1·306	1·688	2·028	2·434	2·719	3·333	3·582
38	0·2551	0·5288	0·6810	0·8512	1·051	1·304	1·686	2·024	2·429	2·712	3·319	3·566
40	0·2550	0·5286	0·6807	0·8507	1·050	1·303	1·684	2·021	2·423	2·704	3·307	3·551
50	0·2547	0·5278	0·6794	0·8489	1·047	1·299	1·676	2·009	2·403	2·678	3·261	3·496
60	0·2545	0·5272	0·6786	0·8477	1·045	1·296	1·671	2·000	2·390	2·660	3·232	3·460
120	0·2539	0·5258	0·6765	0·8446	1·041	1·289	1·658	1·980	2·358	2·617	3·160	3·373
∞	0·2533	0·5244	0·6745	0·8416	1·036	1·282	1·645	1·960	2·326	2·576	3·090	3·291

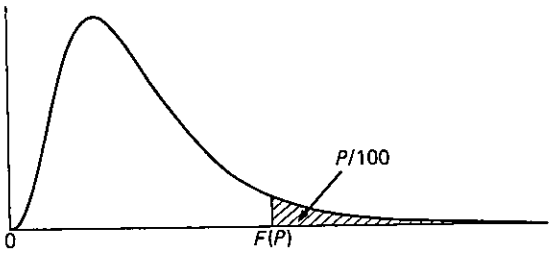
TABLE 12(*a*). 10 PER CENT POINTS OF THE *F*-DISTRIBUTION

The function tabulated is $F(P) = F(P|\nu_1, \nu_2)$ defined by the equation

$$\frac{P}{100} = \frac{\Gamma(\frac{1}{2}\nu_1+\frac{1}{2}\nu_2)}{\Gamma(\frac{1}{2}\nu_1)\,\Gamma(\frac{1}{2}\nu_2)}\,\nu_1^{\frac{1}{2}\nu_1}\,\nu_2^{\frac{1}{2}\nu_2}\int_{F(P)}^{\infty}\frac{F^{\frac{1}{2}\nu_1-1}}{(\nu_2+\nu_1 F)^{\frac{1}{2}(\nu_1+\nu_2)}}\,dF,$$

for P = 10, 5, 2·5, 1, 0·5 and 0·1. The lower percentage points, that is the values $F'(P) = F'(P|\nu_1, \nu_2)$ such that the probability that $F \leqslant F'(P)$ is equal to $P/100$, may be found by the formula

$$F'(P|\nu_1, \nu_2) = 1/F(P|\nu_2, \nu_1).$$

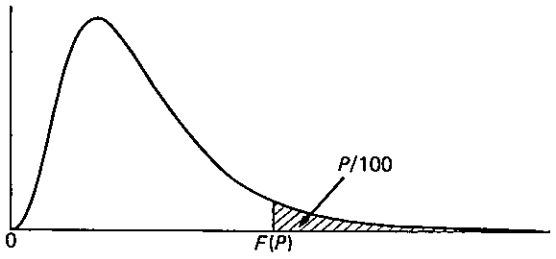


(This shape applies only when $\nu_1 \geqslant 3$. When $\nu_1 < 3$ the mode is at the origin.)

ν_1 =	1	2	3	4	5	6	7	8	10	12	24	∞
ν_2 = 1	39·86	49·50	53·59	55·83	57·24	58·20	58·91	59·44	60·19	60·71	62·00	63·33
2	8·526	9·000	9·162	9·243	9·293	9·326	9·349	9·367	9·392	9·408	9·450	9·491
3	5·538	5·462	5·391	5·343	5·309	5·285	5·266	5·252	5·230	5·216	5·176	5·134
4	4·545	4·325	4·191	4·107	4·051	4·010	3·979	3·955	3·920	3·896	3·831	3·761
5	4·060	3·780	3·619	3·520	3·453	3·405	3·368	3·339	3·297	3·268	3·191	3·105
6	3·776	3·463	3·289	3·181	3·108	3·055	3·014	2·983	2·937	2·905	2·818	2·722
7	3·589	3·257	3·074	2·961	2·883	2·827	2·785	2·752	2·703	2·668	2·575	2·471
8	3·458	3·113	2·924	2·806	2·726	2·668	2·624	2·589	2·538	2·502	2·404	2·293
9	3·360	3·006	2·813	2·693	2·611	2·551	2·505	2·469	2·416	2·379	2·277	2·159
10	3·285	2·924	2·728	2·605	2·522	2·461	2·414	2·377	2·323	2·284	2·178	2·055
11	3·225	2·860	2·660	2·536	2·451	2·389	2·342	2·304	2·248	2·209	2·100	1·972
12	3·177	2·807	2·606	2·480	2·394	2·331	2·283	2·245	2·188	2·147	2·036	1·904
13	3·136	2·763	2·560	2·434	2·347	2·283	2·234	2·195	2·138	2·097	1·983	1·846
14	3·102	2·726	2·522	2·395	2·307	2·243	2·193	2·154	2·095	2·054	1·938	1·797
15	3·073	2·695	2·490	2·361	2·273	2·208	2·158	2·119	2·059	2·017	1·899	1·755
16	3·048	2·668	2·462	2·333	2·244	2·178	2·128	2·088	2·028	1·985	1·866	1·718
17	3·026	2·645	2·437	2·308	2·218	2·152	2·102	2·061	2·001	1·958	1·836	1·686
18	3·007	2·624	2·416	2·286	2·196	2·130	2·079	2·038	1·977	1·933	1·810	1·657
19	2·990	2·606	2·397	2·266	2·176	2·109	2·058	2·017	1·956	1·912	1·787	1·631
20	2·975	2·589	2·380	2·249	2·158	2·091	2·040	1·999	1·937	1·892	1·767	1·607
21	2·961	2·575	2·365	2·233	2·142	2·075	2·023	1·982	1·920	1·875	1·748	1·586
22	2·949	2·561	2·351	2·219	2·128	2·060	2·008	1·967	1·904	1·859	1·731	1·567
23	2·937	2·549	2·339	2·207	2·115	2·047	1·995	1·953	1·890	1·845	1·716	1·549
24	2·927	2·538	2·327	2·195	2·103	2·035	1·983	1·941	1·877	1·832	1·702	1·533
25	2·918	2·528	2·317	2·184	2·092	2·024	1·971	1·929	1·866	1·820	1·689	1·518
26	2·909	2·519	2·307	2·174	2·082	2·014	1·961	1·919	1·855	1·809	1·677	1·504
27	2·901	2·511	2·299	2·165	2·073	2·005	1·952	1·909	1·845	1·799	1·666	1·491
28	2·894	2·503	2·291	2·157	2·064	1·996	1·943	1·900	1·836	1·790	1·656	1·478
29	2·887	2·495	2·283	2·149	2·057	1·988	1·935	1·892	1·827	1·781	1·647	1·467
30	2·881	2·489	2·276	2·142	2·049	1·980	1·927	1·884	1·819	1·773	1·638	1·456
32	2·869	2·477	2·263	2·129	2·036	1·967	1·913	1·870	1·805	1·758	1·622	1·437
34	2·859	2·466	2·252	2·118	2·024	1·955	1·901	1·858	1·793	1·745	1·608	1·419
36	2·850	2·456	2·243	2·108	2·014	1·945	1·891	1·847	1·781	1·734	1·595	1·404
38	2·842	2·448	2·234	2·099	2·005	1·935	1·881	1·838	1·772	1·724	1·584	1·390
40	2·835	2·440	2·226	2·091	1·997	1·927	1·873	1·829	1·763	1·715	1·574	1·377
60	2·791	2·393	2·177	2·041	1·946	1·875	1·819	1·775	1·707	1·657	1·511	1·291
120	2·748	2·347	2·130	1·992	1·896	1·824	1·767	1·722	1·652	1·601	1·447	1·193
∞	2·706	2·303	2·084	1·945	1·847	1·774	1·717	1·670	1·599	1·546	1·383	1·000

TABLE 12(*b*). 5 PER CENT POINTS OF THE *F*-DISTRIBUTION

If $F = \frac{X_1}{\nu_1} \Big/ \frac{X_2}{\nu_2}$, where X_1 and X_2 are independent random variables distributed as χ^2 with ν_1 and ν_2 degrees of freedom respectively, then the probabilities that $F \geqslant F(P)$ and that $F \leqslant F'(P)$ are both equal to $P/100$. Linear interpolation in ν_1 and ν_2 will generally be sufficiently accurate except when either $\nu_1 > 12$ or $\nu_2 > 40$, when harmonic interpolation should be used.



(This shape applies only when $\nu_1 \geqslant 3$. When $\nu_1 < 3$ the mode is at the origin.)

$\nu_1 =$	1	2	3	4	5	6	7	8	10	12	24	∞
$\nu_2 = 1$	161·4	199·5	215·7	224·6	230·2	234·0	236·8	238·9	241·9	243·9	249·1	254·3
2	18·51	19·00	19·16	19·25	19·30	19·33	19·35	19·37	19·40	19·41	19·45	19·50
3	10·13	9·552	9·277	9·117	9·013	8·941	8·887	8·845	8·786	8·745	8·639	8·526
4	7·709	6·944	6·591	6·388	6·256	6·163	6·094	6·041	5·964	5·912	5·774	5·628
5	6·608	5·786	5·409	5·192	5·050	4·950	4·876	4·818	4·735	4·678	4·527	4·365
6	5·987	5·143	4·757	4·534	4·387	4·284	4·207	4·147	4·060	4·000	3·841	3·669
7	5·591	4·737	4·347	4·120	3·972	3·866	3·787	3·726	3·637	3·575	3·410	3·230
8	5·318	4·459	4·066	3·838	3·687	3·581	3·500	3·438	3·347	3·284	3·115	2·928
9	5·117	4·256	3·863	3·633	3·482	3·374	3·293	3·230	3·137	3·073	2·900	2·707
10	4·965	4·103	3·708	3·478	3·326	3·217	3·135	3·072	2·978	2·913	2·737	2·538
11	4·844	3·982	3·587	3·357	3·204	3·095	3·012	2·948	2·854	2·788	2·609	2·404
12	4·747	3·885	3·490	3·259	3·106	2·996	2·913	2·849	2·753	2·687	2·505	2·296
13	4·667	3·806	3·411	3·179	3·025	2·915	2·832	2·767	2·671	2·604	2·420	2·206
14	4·600	3·739	3·344	3·112	2·958	2·848	2·764	2·699	2·602	2·534	2·349	2·131
15	4·543	3·682	3·287	3·056	2·901	2·790	2·707	2·641	2·544	2·475	2·288	2·066
16	4·494	3·634	3·239	3·007	2·852	2·741	2·657	2·591	2·494	2·425	2·235	2·010
17	4·451	3·592	3·197	2·965	2·810	2·699	2·614	2·548	2·450	2·381	2·190	1·960
18	4·414	3·555	3·160	2·928	2·773	2·661	2·577	2·510	2·412	2·342	2·150	1·917
19	4·381	3·522	3·127	2·895	2·740	2·628	2·544	2·477	2·378	2·308	2·114	1·878
20	4·351	3·493	3·098	2·866	2·711	2·599	2·514	2·447	2·348	2·278	2·082	1·843
21	4·325	3·467	3·072	2·840	2·685	2·573	2·488	2·420	2·321	2·250	2·054	1·812
22	4·301	3·443	3·049	2·817	2·661	2·549	2·464	2·397	2·297	2·226	2·028	1·783
23	4·279	3·422	3·028	2·796	2·640	2·528	2·442	2·375	2·275	2·204	2·005	1·757
24	4·260	3·403	3·009	2·776	2·621	2·508	2·423	2·355	2·255	2·183	1·984	1·733
25	4·242	3·385	2·991	2·759	2·603	2·490	2·405	2·337	2·236	2·165	1·964	1·711
26	4·225	3·369	2·975	2·743	2·587	2·474	2·388	2·321	2·220	2·148	1·946	1·691
27	4·210	3·354	2·960	2·728	2·572	2·459	2·373	2·305	2·204	2·132	1·930	1·672
28	4·196	3·340	2·947	2·714	2·558	2·445	2·359	2·291	2·190	2·118	1·915	1·654
29	4·183	3·328	2·934	2·701	2·545	2·432	2·346	2·278	2·177	2·104	1·901	1·638
30	4·171	3·316	2·922	2·690	2·534	2·421	2·334	2·266	2·165	2·092	1·887	1·622
32	4·149	3·295	2·901	2·668	2·512	2·399	2·313	2·244	2·142	2·070	1·864	1·594
34	4·130	3·276	2·883	2·650	2·494	2·380	2·294	2·225	2·123	2·050	1·843	1·569
36	4·113	3·259	2·866	2·634	2·477	2·364	2·277	2·209	2·106	2·033	1·824	1·547
38	4·098	3·245	2·852	2·619	2·463	2·349	2·262	2·194	2·091	2·017	1·808	1·527
40	4·085	3·232	2·839	2·606	2·449	2·336	2·249	2·180	2·077	2·003	1·793	1·509
60	4·001	3·150	2·758	2·525	2·368	2·254	2·167	2·097	1·993	1·917	1·700	1·389
120	3·920	3·072	2·680	2·447	2·290	2·175	2·087	2·016	1·910	1·834	1·608	1·254
∞	3·841	2·996	2·605	2·372	2·214	2·099	2·010	1·938	1·831	1·752	1·517	1·000

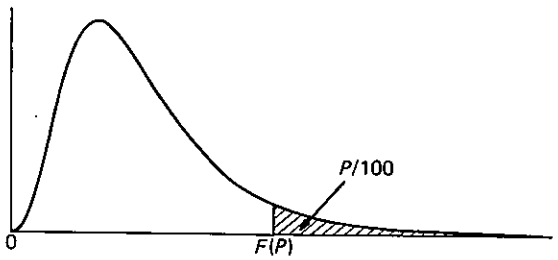
TABLE 12(*c*). 2·5 PER CENT POINTS OF THE *F*-DISTRIBUTION

The function tabulated is $F(P) = F(P|\nu_1, \nu_2)$ defined by the equation

$$\frac{P}{100} = \frac{\Gamma(\frac{1}{2}\nu_1+\frac{1}{2}\nu_2)}{\Gamma(\frac{1}{2}\nu_1)\,\Gamma(\frac{1}{2}\nu_2)}\,\nu_1^{\frac{1}{2}\nu_1}\,\nu_2^{\frac{1}{2}\nu_2}\int_{F(P)}^{\infty}\frac{F^{\frac{1}{2}\nu_1-1}}{(\nu_2+\nu_1 F)^{\frac{1}{2}(\nu_1+\nu_2)}}\,dF,$$

for P = 10, 5, 2·5, 1, 0·5 and 0·1. The lower percentage points, that is the values $F'(P) = F'(P|\nu_1, \nu_2)$ such that the probability that $F \leqslant F'(P)$ is equal to $P/100$, may be found by the formula

$$F'(P|\nu_1, \nu_2) = 1/F(P|\nu_2, \nu_1).$$

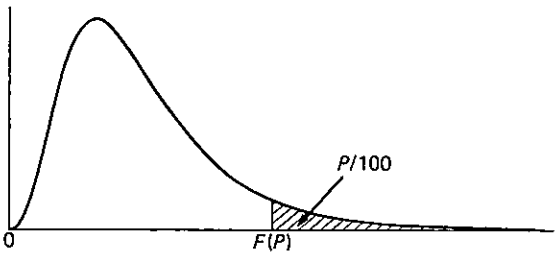


(This shape applies only when $\nu_1 \geqslant 3$. When $\nu_1 < 3$ the mode is at the origin.)

ν_1 =	1	2	3	4	5	6	7	8	10	12	24	∞
ν_2 = 1	647·8	799·5	864·2	899·6	921·8	937·1	948·2	956·7	968·6	976·7	997·2	1018
2	38·51	39·00	39·17	39·25	39·30	39·33	39·36	39·37	39·40	39·41	39·46	39·50
3	17·44	16·04	15·44	15·10	14·88	14·73	14·62	14·54	14·42	14·34	14·12	13·90
4	12·22	10·65	9·979	9·605	9·364	9·197	9·074	8·980	8·844	8·751	8·511	8·257
5	10·01	8·434	7·764	7·388	7·146	6·978	6·853	6·757	6·619	6·525	6·278	6·015
6	8·813	7·260	6·599	6·227	5·988	5·820	5·695	5·600	5·461	5·366	5·117	4·849
7	8·073	6·542	5·890	5·523	5·285	5·119	4·995	4·899	4·761	4·666	4·415	4·142
8	7·571	6·059	5·416	5·053	4·817	4·652	4·529	4·433	4·295	4·200	3·947	3·670
9	7·209	5·715	5·078	4·718	4·484	4·320	4·197	4·102	3·964	3·868	3·614	3·333
10	6·937	5·456	4·826	4·468	4·236	4·072	3·950	3·855	3·717	3·621	3·365	3·080
11	6·724	5·256	4·630	4·275	4·044	3·881	3·759	3·664	3·526	3·430	3·173	2·883
12	6·554	5·096	4·474	4·121	3·891	3·728	3·607	3·512	3·374	3·277	3·019	2·725
13	6·414	4·965	4·347	3·996	3·767	3·604	3·483	3·388	3·250	3·153	2·893	2·595
14	6·298	4·857	4·242	3·892	3·663	3·501	3·380	3·285	3·147	3·050	2·789	2·487
15	6·200	4·765	4·153	3·804	3·576	3·415	3·293	3·199	3·060	2·963	2·701	2·395
16	6·115	4·687	4·077	3·729	3·502	3·341	3·219	3·125	2·986	2·889	2·625	2·316
17	6·042	4·619	4·011	3·665	3·438	3·277	3·156	3·061	2·922	2·825	2·560	2·247
18	5·978	4·560	3·954	3·608	3·382	3·221	3·100	3·005	2·866	2·769	2·503	2·187
19	5·922	4·508	3·903	3·559	3·333	3·172	3·051	2·956	2·817	2·720	2·452	2·133
20	5·871	4·461	3·859	3·515	3·289	3·128	3·007	2·913	2·774	2·676	2·408	2·085
21	5·827	4·420	3·819	3·475	3·250	3·090	2·969	2·874	2·735	2·637	2·368	2·042
22	5·786	4·383	3·783	3·440	3·215	3·055	2·934	2·839	2·700	2·602	2·331	2·003
23	5·750	4·349	3·750	3·408	3·183	3·023	2·902	2·808	2·668	2·570	2·299	1·968
24	5·717	4·319	3·721	3·379	3·155	2·995	2·874	2·779	2·640	2·541	2·269	1·935
25	5·686	4·291	3·694	3·353	3·129	2·969	2·848	2·753	2·613	2·515	2·242	1·906
26	5·659	4·265	3·670	3·329	3·105	2·945	2·824	2·729	2·590	2·491	2·217	1·878
27	5·633	4·242	3·647	3·307	3·083	2·923	2·802	2·707	2·568	2·469	2·195	1·853
28	5·610	4·221	3·626	3·286	3·063	2·903	2·782	2·687	2·547	2·448	2·174	1·829
29	5·588	4·201	3·607	3·267	3·044	2·884	2·763	2·669	2·529	2·430	2·154	1·807
30	5·568	4·182	3·589	3·250	3·026	2·867	2·746	2·651	2·511	2·412	2·136	1·787
32	5·531	4·149	3·557	3·218	2·995	2·836	2·715	2·620	2·480	2·381	2·103	1·750
34	5·499	4·120	3·529	3·191	2·968	2·808	2·688	2·593	2·453	2·353	2·075	1·717
36	5·471	4·094	3·505	3·167	2·944	2·785	2·664	2·569	2·429	2·329	2·049	1·687
38	5·446	4·071	3·483	3·145	2·923	2·763	2·643	2·548	2·407	2·307	2·027	1·661
40	5·424	4·051	3·463	3·126	2·904	2·744	2·624	2·529	2·388	2·288	2·007	1·637
60	5·286	3·925	3·343	3·008	2·786	2·627	2·507	2·412	2·270	2·169	1·882	1·482
120	5·152	3·805	3·227	2·894	2·674	2·515	2·395	2·299	2·157	2·055	1·760	1·310
∞	5·024	3·689	3·116	2·786	2·567	2·408	2·288	2·192	2·048	1·945	1·640	1·000

TABLE 12(*d*). 1 PER CENT POINTS OF THE *F*-DISTRIBUTION

If $F = \frac{X_1}{\nu_1} \Big/ \frac{X_2}{\nu_2}$, where X_1 and X_2 are independent random variables distributed as χ^2 with ν_1 and ν_2 degrees of freedom respectively, then the probabilities that $F \geqslant F(P)$ and that $F \leqslant F'(P)$ are both equal to $P/100$. Linear interpolation in ν_1 or ν_2 will generally be sufficiently accurate except when either $\nu_1 > 12$ or $\nu_2 > 40$, when harmonic interpolation should be used.



(This shape applies only when $\nu_1 \geqslant 3$. When $\nu_1 < 3$ the mode is at the origin.)

$\nu_1 =$	1	2	3	4	5	6	7	8	10	12	24	∞
$\nu_2 = 1$	4052	4999	5403	5625	5764	5859	5928	5981	6056	6106	6235	6366
2	98·50	99·00	99·17	99·25	99·30	99·33	99·36	99·37	99·40	99·42	99·46	99·50
3	34·12	30·82	29·46	28·71	28·24	27·91	27·67	27·49	27·23	27·05	26·60	26·13
4	21·20	18·00	16·69	15·98	15·52	15·21	14·98	14·80	14·55	14·37	13·93	13·46
5	16·26	13·27	12·06	11·39	10·97	10·67	10·46	10·29	10·05	9·888	9·466	9·020
6	13·75	10·92	9·780	9·148	8·746	8·466	8·260	8·102	7·874	7·718	7·313	6·880
7	12·25	9·547	8·451	7·847	7·460	7·191	6·993	6·840	6·620	6·469	6·074	5·650
8	11·26	8·649	7·591	7·006	6·632	6·371	6·178	6·029	5·814	5·667	5·279	4·859
9	10·56	8·022	6·992	6·422	6·057	5·802	5·613	5·467	5·257	5·111	4·729	4·311
10	10·04	7·559	6·552	5·994	5·636	5·386	5·200	5·057	4·849	4·706	4·327	3·909
11	9·646	7·206	6·217	5·668	5·316	5·069	4·886	4·744	4·539	4·397	4·021	3·602
12	9·330	6·927	5·953	5·412	5·064	4·821	4·640	4·499	4·296	4·155	3·780	3·361
13	9·074	6·701	5·739	5·205	4·862	4·620	4·441	4·302	4·100	3·960	3·587	3·165
14	8·862	6·515	5·564	5·035	4·695	4·456	4·278	4·140	3·939	3·800	3·427	3·004
15	8·683	6·359	5·417	4·893	4·556	4·318	4·142	4·004	3·805	3·666	3·294	2·868
16	8·531	6·226	5·292	4·773	4·437	4·202	4·026	3·890	3·691	3·553	3·181	2·753
17	8·400	6·112	5·185	4·669	4·336	4·102	3·927	3·791	3·593	3·455	3·084	2·653
18	8·285	6·013	5·092	4·579	4·248	4·015	3·841	3·705	3·508	3·371	2·999	2·566
19	8·185	5·926	5·010	4·500	4·171	3·939	3·765	3·631	3·434	3·297	2·925	2·489
20	8·096	5·849	4·938	4·431	4·103	3·871	3·699	3·564	3·368	3·231	2·859	2·421
21	8·017	5·780	4·874	4·369	4·042	3·812	3·640	3·506	3·310	3·173	2·801	2·360
22	7·945	5·719	4·817	4·313	3·988	3·758	3·587	3·453	3·258	3·121	2·749	2·305
23	7·881	5·664	4·765	4·264	3·939	3·710	3·539	3·406	3·211	3·074	2·702	2·256
24	7·823	5·614	4·718	4·218	3·895	3·667	3·496	3·363	3·168	3·032	2·659	2·211
25	7·770	5·568	4·675	4·177	3·855	3·627	3·457	3·324	3·129	2·993	2·620	2·169
26	7·721	5·526	4·637	4·140	3·818	3·591	3·421	3·288	3·094	2·958	2·585	2·131
27	7·677	5·488	4·601	4·106	3·785	3·558	3·388	3·256	3·062	2·926	2·552	2·097
28	7·636	5·453	4·568	4·074	3·754	3·528	3·358	3·226	3·032	2·896	2·522	2·064
29	7·598	5·420	4·538	4·045	3·725	3·499	3·330	3·198	3·005	2·868	2·495	2·034
30	7·562	5·390	4·510	4·018	3·699	3·473	3·304	3·173	2·979	2·843	2·469	2·006
32	7·499	5·336	4·459	3·969	3·652	3·427	3·258	3·127	2·934	2·798	2·423	1·956
34	7·444	5·289	4·416	3·927	3·611	3·386	3·218	3·087	2·894	2·758	2·383	1·911
36	7·396	5·248	4·377	3·890	3·574	3·351	3·183	3·052	2·859	2·723	2·347	1·872
38	7·353	5·211	4·343	3·858	3·542	3·319	3·152	3·021	2·828	2·692	2·316	1·837
40	7·314	5·179	4·313	3·828	3·514	3·291	3·124	2·993	2·801	2·665	2·288	1·805
60	7·077	4·977	4·126	3·649	3·339	3·119	2·953	2·823	2·632	2·496	2·115	1·601
120	6·851	4·787	3·949	3·480	3·174	2·956	2·792	2·663	2·472	2·336	1·950	1·381
∞	6·635	4·605	3·782	3·319	3·017	2·802	2·639	2·511	2·321	2·185	1·791	1·000

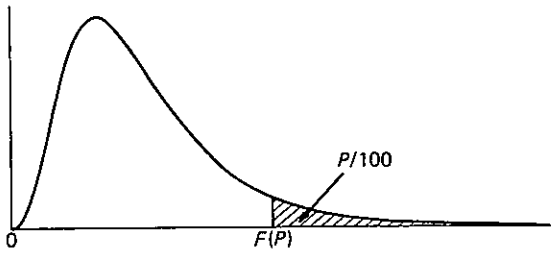
TABLE 12(*e*). 0·5 PER CENT POINTS OF THE *F*-DISTRIBUTION

The function tabulated is $F(P) = F(P|\nu_1, \nu_2)$ defined by the equation

$$\frac{P}{100} = \frac{\Gamma(\frac{1}{2}\nu_1 + \frac{1}{2}\nu_2)}{\Gamma(\frac{1}{2}\nu_1)\,\Gamma(\frac{1}{2}\nu_2)}\,\nu_1^{\frac{1}{2}\nu_1}\,\nu_2^{\frac{1}{2}\nu_2}\int_{F(P)}^{\infty}\frac{F^{\frac{1}{2}\nu_1 - 1}}{(\nu_2 + \nu_1 F)^{\frac{1}{2}(\nu_1+\nu_2)}}\,dF,$$

for P = 10, 5, 2·5, 1, 0·5 and 0·1. The lower percentage points, that is the values $F'(P) = F'(P|\nu_1, \nu_2)$ such that the probability that $F \leqslant F'(P)$ is equal to $P/100$, may be found by the formula

$$F'(P|\nu_1, \nu_2) = 1/F(P|\nu_2, \nu_1).$$

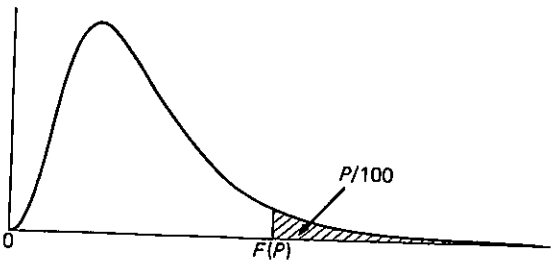


(This shape applies only when $\nu_1 \geqslant 3$. When $\nu_1 < 3$ the mode is at the origin.)

ν_1 =	1	2	3	4	5	6	7	8	10	12	24	∞
ν_2 = 1	16211	20000	21615	22500	23056	23437	23715	23925	24224	24426	24940	25464
2	198·5	199·0	199·2	199·2	199·3	199·3	199·4	199·4	199·4	199·4	199·5	199·5
3	55·55	49·80	47·47	46·19	45·39	44·84	44·43	44·13	43·69	43·39	42·62	41·83
4	31·33	26·28	24·26	23·15	22·46	21·97	21·62	21·35	20·97	20·70	20·03	19·32
5	22·78	18·31	16·53	15·56	14·94	14·51	14·20	13·96	13·62	13·38	12·78	12·14
6	18·63	14·54	12·92	12·03	11·46	11·07	10·79	10·57	10·25	10·03	9·474	8·879
7	16·24	12·40	10·88	10·05	9·522	9·155	8·885	8·678	8·380	8·176	7·645	7·076
8	14·69	11·04	9·596	8·805	8·302	7·952	7·694	7·496	7·211	7·015	6·503	5·951
9	13·61	10·11	8·717	7·956	7·471	7·134	6·885	6·693	6·417	6·227	5·729	5·188
10	12·83	9·427	8·081	7·343	6·872	6·545	6·302	6·116	5·847	5·661	5·173	4·639
11	12·23	8·912	7·600	6·881	6·422	6·102	5·865	5·682	5·418	5·236	4·756	4·226
12	11·75	8·510	7·226	6·521	6·071	5·757	5·525	5·345	5·085	4·906	4·431	3·904
13	11·37	8·186	6·926	6·233	5·791	5·482	5·253	5·076	4·820	4·643	4·173	3·647
14	11·06	7·922	6·680	5·998	5·562	5·257	5·031	4·857	4·603	4·428	3·961	3·436
15	10·80	7·701	6·476	5·803	5·372	5·071	4·847	4·674	4·424	4·250	3·786	3·260
16	10·58	7·514	6·303	5·638	5·212	4·913	4·692	4·521	4·272	4·099	3·638	3·112
17	10·38	7·354	6·156	5·497	5·075	4·779	4·559	4·389	4·142	3·971	3·511	2·984
18	10·22	7·215	6·028	5·375	4·956	4·663	4·445	4·276	4·030	3·860	3·402	2·873
19	10·07	7·093	5·916	5·268	4·853	4·561	4·345	4·177	3·933	3·763	3·306	2·776
20	9·944	6·986	5·818	5·174	4·762	4·472	4·257	4·090	3·847	3·678	3·222	2·690
21	9·830	6·891	5·730	5·091	4·681	4·393	4·179	4·013	3·771	3·602	3·147	2·614
22	9·727	6·806	5·652	5·017	4·609	4·322	4·109	3·944	3·703	3·535	3·081	2·545
23	9·635	6·730	5·582	4·950	4·544	4·259	4·047	3·882	3·642	3·475	3·021	2·484
24	9·551	6·661	5·519	4·890	4·486	4·202	3·991	3·826	3·587	3·420	2·967	2·428
25	9·475	6·598	5·462	4·835	4·433	4·150	3·939	3·776	3·537	3·370	2·918	2·377
26	9·406	6·541	5·409	4·785	4·384	4·103	3·893	3·730	3·492	3·325	2·873	2·330
27	9·342	6·489	5·361	4·740	4·340	4·059	3·850	3·687	3·450	3·284	2·832	2·287
28	9·284	6·440	5·317	4·698	4·300	4·020	3·811	3·649	3·412	3·246	2·794	2·247
29	9·230	6·396	5·276	4·659	4·262	3·983	3·775	3·613	3·377	3·211	2·759	2·210
30	9·180	6·355	5·239	4·623	4·228	3·949	3·742	3·580	3·344	3·179	2·727	2·176
32	9·090	6·281	5·171	4·559	4·166	3·889	3·682	3·521	3·286	3·121	2·670	2·114
34	9·012	6·217	5·113	4·504	4·112	3·836	3·630	3·470	3·235	3·071	2·620	2·060
36	8·943	6·161	5·062	4·455	4·065	3·790	3·585	3·425	3·191	3·027	2·576	2·013
38	8·882	6·111	5·016	4·412	4·023	3·749	3·545	3·385	3·152	2·988	2·537	1·970
40	8·828	6·066	4·976	4·374	3·986	3·713	3·509	3·350	3·117	2·953	2·502	1·932
60	8·495	5·795	4·729	4·140	3·760	3·492	3·291	3·134	2·904	2·742	2·290	1·689
120	8·179	5·539	4·497	3·921	3·548	3·285	3·087	2·933	2·705	2·544	2·089	1·431
∞	7·879	5·298	4·279	3·715	3·350	3·091	2·897	2·744	2·519	2·358	1·898	1·000

TABLE 12(f). 0·1 PER CENT POINTS OF THE F-DISTRIBUTION

If $F = \frac{X_1}{\nu_1} \Big/ \frac{X_2}{\nu_2}$, where X_1 and X_2 are independent random variables distributed as χ^2 with ν_1 and ν_2 degrees of freedom respectively, then the probabilities that $F \geqslant F(P)$ and that $F \leqslant F'(P)$ are both equal to $P/100$. Linear interpolation in ν_1 or ν_2 will generally be sufficiently accurate except when either $\nu_1 > 12$ or $\nu_2 > 40$, when harmonic interpolation should be used.



(This shape applies only when $\nu_1 \geqslant 3$. When $\nu_1 < 3$ the mode is at the origin.)

ν_2 =	ν_1 = 1	2	3	4	5	6	7	8	10	12	24	∞
1*	4053	5000	5404	5625	5764	5859	5929	5981	6056	6107	6235	6366
2	998·5	999·0	999·2	999·2	999·3	999·3	999·4	999·4	999·4	999·4	999·5	999·5
3	167·0	148·5	141·1	137·1	134·6	132·8	131·6	130·6	129·2	128·3	125·9	123·5
4	74·14	61·25	56·18	53·44	51·71	50·53	49·66	49·00	48·05	47·41	45·77	44·05
5	47·18	37·12	33·20	31·09	29·75	28·83	28·16	27·65	26·92	26·42	25·13	23·79
6	35·51	27·00	23·70	21·92	20·80	20·03	19·46	19·03	18·41	17·99	16·90	15·75
7	29·25	21·69	18·77	17·20	16·21	15·52	15·02	14·63	14·08	13·71	12·73	11·70
8	25·41	18·49	15·83	14·39	13·48	12·86	12·40	12·05	11·54	11·19	10·30	9·334
9	22·86	16·39	13·90	12·56	11·71	11·13	10·70	10·37	9·894	9·570	8·724	7·813
10	21·04	14·91	12·55	11·28	10·48	9·926	9·517	9·204	8·754	8·445	7·638	6·762
11	19·69	13·81	11·56	10·35	9·578	9·047	8·655	8·355	7·922	7·626	6·847	5·998
12	18·64	12·97	10·80	9·633	8·892	8·379	8·001	7·710	7·292	7·005	6·249	5·420
13	17.82	12·31	10·21	9·073	8·354	7·856	7·489	7·206	6·799	6·519	5·781	4·967
14	17·14	11·78	9·729	8·622	7·922	7·436	7·077	6·802	6·404	6·130	5·407	4·604
15	16·59	11·34	9·335	8·253	7·567	7·092	6·741	6·471	6·081	5·812	5·101	4·307
16	16·12	10·97	9·006	7·944	7·272	6·805	6·460	6·195	5·812	5·547	4·846	4·059
17	15·72	10·66	8·727	7·683	7·022	6·562	6·223	5·962	5·584	5·324	4·631	3·850
18	15·38	10·39	8·487	7·459	6·808	6·355	6·021	5·763	5·390	5·132	4·447	3·670
19	15·08	10·16	8·280	7·265	6·622	6·175	5·845	5·590	5·222	4·967	4·288	3·514
20	14·82	9·953	8·098	7·096	6·461	6·019	5·692	5·440	5·075	4·823	4·149	3·378
21	14·59	9·772	7·938	6·947	6·318	5·881	5·557	5·308	4·946	4·696	4·027	3·257
22	14·38	9·612	7·796	6·814	6·191	5·758	5·438	5·190	4·832	4·583	3·919	3·151
23	14·20	9·469	7·669	6·696	6·078	5·649	5·331	5·085	4·730	4·483	3·822	3·055
24	14·03	9·339	7·554	6·589	5·977	5·550	5·235	4·991	4·638	4·393	3·735	2·969
25	13·88	9·223	7·451	6·493	5·885	5·462	5·148	4·906	4·555	4·312	3·657	2·890
26	13·74	9·116	7·357	6·406	5·802	5·381	5·070	4·829	4·480	4·238	3·586	2·819
27	13·61	9·019	7·272	6·326	5·726	5·308	4·998	4·759	4·412	4·171	3·521	2·754
28	13·50	8·931	7·193	6·253	5·656	5·241	4·933	4·695	4·349	4·109	3·462	2·695
29	13·39	8·849	7·121	6·186	5·593	5·179	4·873	4·636	4·292	4·053	3·407	2·640
30	13·29	8·773	7·054	6·125	5·534	5·122	4·817	4·581	4·239	4·001	3·357	2·589
32	13·12	8·639	6·936	6·014	5·429	5·021	4·719	4·485	4·145	3·908	3·268	2·498
34	12·97	8·522	6·833	5·919	5·339	4·934	4·633	4·401	4·063	3·828	3·191	2·419
36	12·83	8·420	6·744	5·836	5·260	4·857	4·559	4·328	3·992	3·758	3·123	2·349
38	12·71	8·331	6·665	5·763	5·190	4·790	4·494	4·264	3·930	3·697	3·064	2·288
40	12·61	8·251	6·595	5·698	5·128	4·731	4·436	4·207	3·874	3·642	3·011	2·233
60	11·97	7·768	6·171	5·307	4·757	4·372	4·086	3·865	3·541	3·315	2·694	1·890
120	11·38	7·321	5·781	4·947	4·416	4·044	3·767	3·552	3·237	3·016	2·402	1·543
∞	10·83	6·908	5·422	4·617	4·103	3·743	3·475	3·266	2·959	2·742	2·132	1·000

* Entries in the row $\nu_2 = 1$ must be multiplied by 100.